



Analysis of a comprehensive inventory of rainfall-induced landslides and multi-factor coupling mechanisms in Eastern Guangdong, China, in August 2018

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Received: 18 July 2025 / Accepted: 22 November 2025 / Published online: 5 December 2025
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Abstract

This paper takes the extreme rainfall event in eastern Guangdong in August 2018 as a case study. We systematically compiled an inventory of 6194 rainfall-triggered landslides and combined the AutoGluon framework with SHapley Additive exPlanations (SHAP) analysis to reveal the multi-factor coupling mechanisms governing landslide occurrence. The research identifies low-to-medium elevation (74–374 m), gentle slopes (10–30°), high vegetation coverage (NDVI 210–255), and cumulative rainfall exceeding 380 mm as key characteristics of high-incidence landslide zones. Landslides were particularly pronounced in weathered Jurassic granite formations and within 2000 m of rivers. Landslide distribution is significantly controlled by the synergistic interaction of four dominant factors: elevation, cumulative rainfall, slope angle, and normalized difference vegetation index (NDVI). Binary and ternary factor coupling analyses further indicate that low-to-medium elevation topography enhances local rainfall intensity through the ‘windward slope rainfall amplification effect,’ while the combination of colluvial deposits on gentle slopes and root channels in vegetated areas accelerates rainwater infiltration. This process disrupts the critical equilibrium state of slopes, triggering landslides. The research proposes two ternary threshold frameworks: ‘low-to-medium elevation–gentle slope–intense rainfall’ and ‘low-to-medium elevation–gentle slope–high vegetation coverage’. These frameworks provide a scientific basis for rainfall-induced cluster landslide risk assessment and early warning systems in coastal South China.

Keywords Landslide inventory · Spatial distribution characteristics · Automated machine learning · Multi-Factor coupling mechanisms · SHAP

Introduction

Coastal South China, under the combined influence of monsoon climate and complex topography, is one of China’s most frequent regions for rainfall-triggered landslides (Liu and Chen 2019; Tan et al. 2025). In recent years, against the backdrop of global warming and increasing extreme rainfall events, cluster landslides in this region have exhibited distinct characteristics of high frequency, small scale, and strong spatial clustering (Bai et al. 2021; Feng et al. 2022; Ma et al. 2022; Yang et al. 2025; Zhang et al. 2025a). Although rainfall is the primary triggering factor, the hazard processes are strongly influenced by complex interactions among multiple environmental factors, including terrain morphology, geological conditions, hydrological characteristics, and vegetation cover (Bernardie et al. 2015; Kohler et al. 2023; Li et al. 2010; Tatard et al. 2010). Traditional approaches typically employ

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statistical analyses to correlate influencing factors or rainfall thresholds with landslide distribution patterns, often focusing on single factors. These methods struggle to reveal the complex mechanisms underlying multi-factor interactions (KC et al. 2024; Wang et al. 2007; Xiao et al. 2023). Furthermore, quantitative analysis of nonlinear coupling relationships among multiple factors is frequently inadequate, limiting deeper understanding of disaster mechanisms (Li et al. 2021; Shwe et al. 2024; Zhou et al. 2025). The existing research, although based on various methods such as spatiotemporal convergence cross-mapping (CCM), high displacement areas (AHDs), optimization models, etc., has explored the coupling relationship between landslides and hydrological factors (Ling et al. 2025; Wang et al. 2025; Wei et al. 2025). However, there are still significant gaps in the quantitative analysis of multi-factor synergies and the research of landslide clusters triggered by extreme events, particularly the lack of systematic research on landslide inventory construction and in-depth analysis of factor coupling mechanisms (Jia et al. 2025; Li et al. 2024b; Liu et al. 2025a; Xie et al. 2025d). Current research methods and theoretical frameworks are mostly limited to the single analysis of static factors, failing to effectively reveal the synergistic effects and nonlinear coupling relationships between factors, which limits the early warning and risk assessment capabilities for landslide disasters.

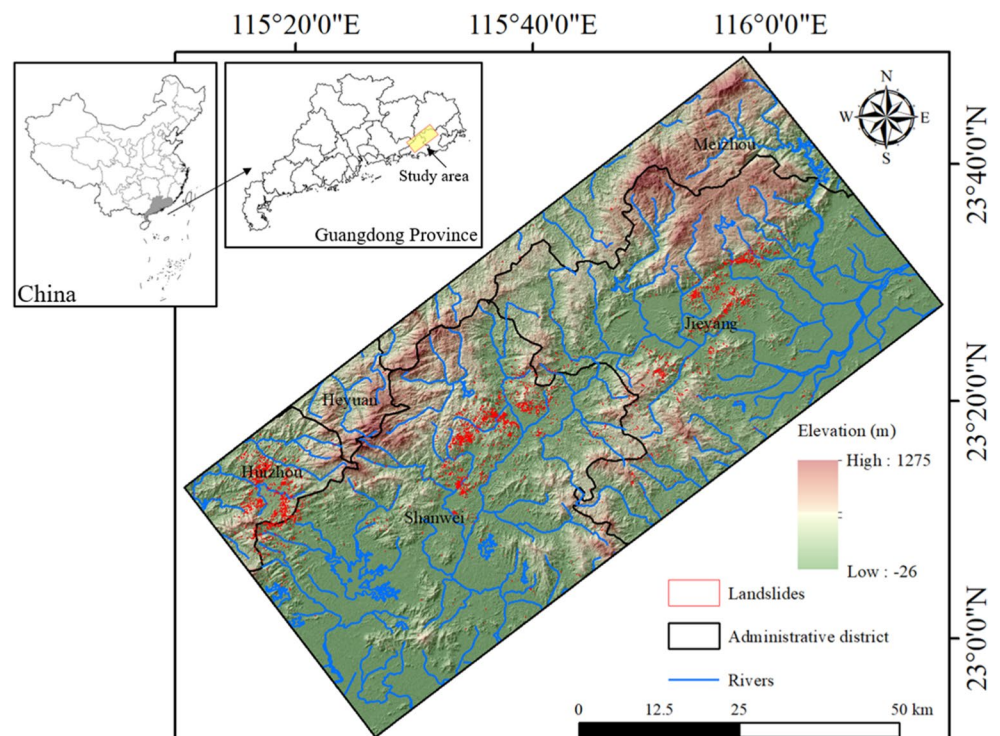
In August 2018, eastern Guangdong experienced an unprecedented extreme rainfall event, triggering thousands of medium- and small-scale landslides. This paper, based on a comprehensive landslide inventory, systematically analyzes

the spatial distribution characteristics of landslides under extreme rainfall events and the coupling mechanisms between landslides and various environmental factors. By applying automated machine learning and interpretability techniques, the study identifies and quantifies the dominant factors influencing landslide occurrence and their relative contributions. Additionally, through binary and ternary factor coupling analysis, a multi-factor nonlinear coupling framework is proposed, revealing the complex interactions and threshold effects between rainfall, topography, geology, hydrology, and other factors. This research provides new theoretical support for landslide risk assessment and early warning.

Study area

The study area is located in eastern Guangdong Province, encompassing five prefecture-level cities. Centered on Jieyang, Heyuan, Huizhou, Shanwei, and Meizhou cities, it covers a significant portion of eastern Guangdong (Fig. 1). The study area boundary was determined by first interpreting landslide-dense zones, then extending 3–5 km outward for verification based on landslide concentration gradients. Geographically, the area spans 115°9'E to 116°14'E longitude and 22°51'N to 23°50'N latitude, with a total area of 5012.41 km². The region features a well-developed river network and exhibits significant topographic relief, sloping northwest to southeast. Elevations range from −26 m to 1275 m, creating pronounced altitudinal variations.

Fig. 1 Geographic location, topographic characteristics, and distribution of rainfall-induced landslides in the study area (administrative boundaries shown)



Dominant landforms include mountains, hills, coastal lowlands, and river valleys/basins. Climatically, the area experiences a subtropical monsoon climate with abundant rainfall. Intense precipitation events primarily occur between April and September, substantially increasing the probability of geological hazards during this period.

Data and methods

Remote sensing data

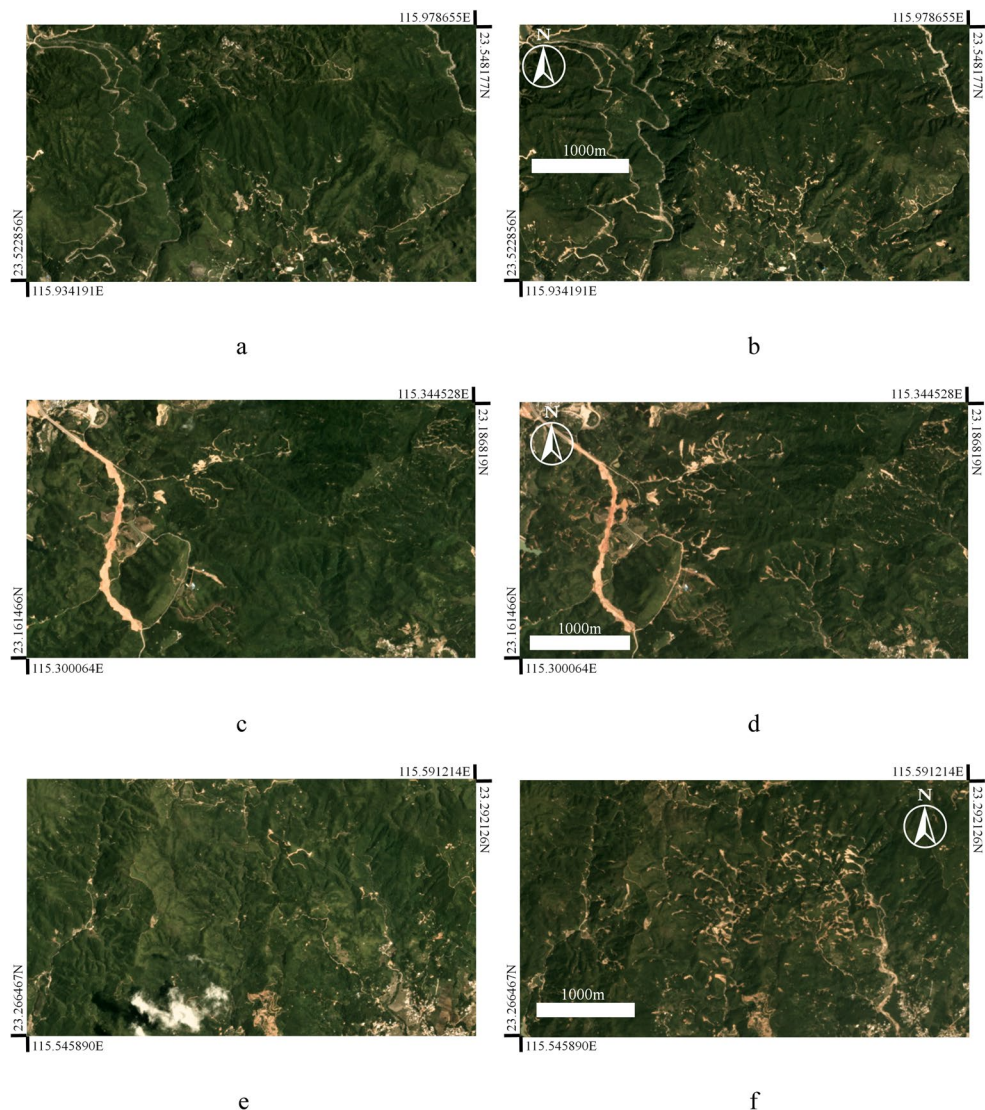
Base imagery comprised 3 m-resolution scenes acquired in mid-August and mid-September 2018, augmented by daily acquisitions. Both the interpretability of the imagery and the temporal baseline meet the requirements for landslide mapping, confirming the feasibility of constructing an event-specific inventory. Inter-comparison of multi-temporal

images ensures that the majority of mapped landslides were initiated during the target rainfall episode. As illustrated in Fig. 2, pre- and post-rainfall images from three representative areas clearly reveal landslide boundaries.

Precipitation data

This paper utilized NASA's Global Precipitation Measurement (GPM) daily rainfall product to analyze the precipitation dynamics of the event. The gridded GPM data (source: <https://search.earthdata.nasa.gov>) covering 92 days from July 1 to September 30, 2018, were processed to characterize rainfall patterns. Eleven sampling points (a-k) within landslide-dense areas were selected for detailed analysis. As shown in Fig. 3b, accumulated rainfall at all 11 points exhibited a steady increasing trend during the monitoring period. A pronounced acceleration occurred exclusively between August 22 and September 2, with point f recording

Fig. 2 Pre- and post-rainfall event satellite imagery of representative landslide-dense areas. (a) Mid-august image; (b) mid-september image; (c) mid-august image; (d) mid-september image; (e) mid-august image; (f) mid-september image



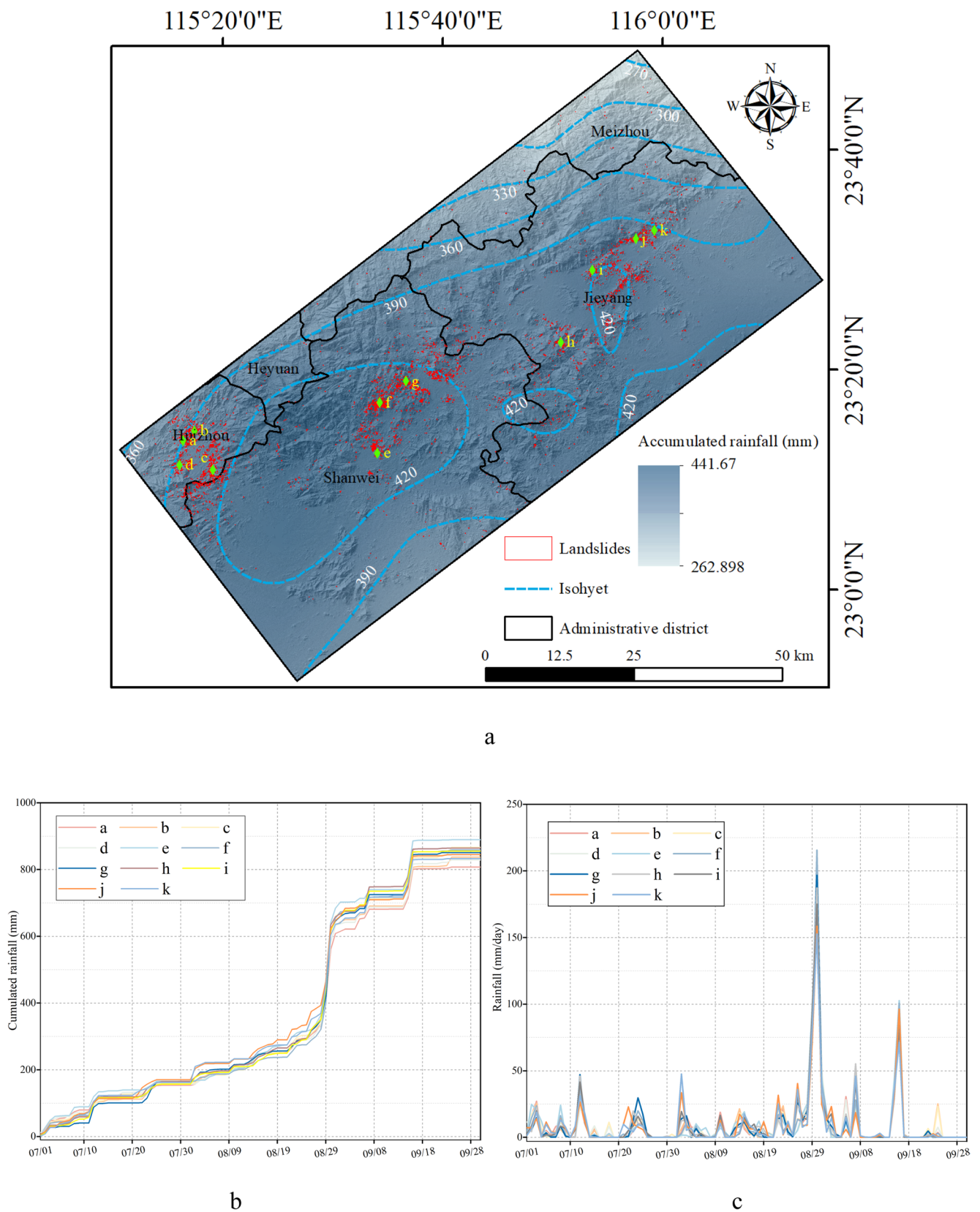


Fig. 3 Rainfall distribution within the study area and detailed precipitation metrics at 11 sampling points. **(a)** Spatial distribution of accumulated rainfall; **(b)** time-series of accumulated rainfall (92 days) across sampling points; **(c)** daily rainfall variation (92 days) depicted via dot plot matrix

the maximum 12-day accumulation (437.68 mm), followed by point i (428.48 mm). The spatial distribution of event-total accumulated rainfall during this critical 12-day period is presented in Fig. 3a. Comparative analysis of daily rainfall across the three-month period (Fig. 3c) reveals August 30 as the most extreme precipitation day, with intensities exceeding other dates by factors of 3–6. Peak daily rainfall reached 215.62 mm at point f and 196.74 mm at point g.

Impact factor data

Rainfall-induced landslides are governed by the combined effects of rainfall amount, topographic-geomorphometric attributes, and hydrogeological conditions. Accordingly, this paper selected fourteen conditioning factors for subsequent analyses: elevation, slope, aspect, topographic relief, slope curvature, profile curvature, plane curvature, lithostratigraphic unit, topographic wetness index (TWI), distance to river, land-cover type, soil type, NDVI, and accumulated rainfall (Fig. 4).

The elevation data were derived from ALOS PALSAR 12.5 m resolution DEM accessed via ASF DAAC (<https://seamless.usf.alaska.edu/#/?dataset=ALOS>). Subsequent terrain parameters—including slope gradient, aspect, topographic relief, slope curvature, profile curvature, plan curvature, and Topographic Wetness Index (TWI)—were generated using the GeoScene geospatial analytics platform. NDVI data originated from the ‘Annual Maximum NDVI Dataset of China (2000–2020) at 30-m Resolution’ provided by the National Ecosystem Science Data Center. This dataset features 30-meter spatial resolution and annual temporal resolution, with the 2018 composite selected for analysis (Yang et al. 2019). Hydrographic data were obtained from the ‘1:1,000,000 Public Edition Fundamental Geographic Information Data (2021)’ through the National Catalog Service for Geographic Information (<https://www.webmap.cn>). River networks within the study area were extracted, with proximity analysis yielding rasterized distance-to-river metrics. Fault lines and lithological data were sourced from the ‘National 1:200,000 Digital Geological Map Spatial Database’ (Li et al. 2019). Original MapGIS format files were processed through feature extraction and format conversion to generate fault and lithology layers. Corresponding lithological classifications for each stratigraphic epoch are detailed in Table 1. Land cover data were derived from China’s first Landsat-based annual land cover product (1985–2019), developed on Google Earth Engine and categorized into nine classes: Cropland, Forest, Shrub, Grassland, Water, Snow/Ice, Barren, Impervious, and Wetland (Yang and Huang 2021). Soil type data originated from a 30-meter resolution spatial distribution dataset generated through digital reconstruction of the 1995 Soil Map of the

People’s Republic of China, published by the National Soil Survey Office. This national-coverage dataset integrates land use patterns with ground-surveyed pedological data. Accumulated rainfall was calculated from NASA’s Global Precipitation Measurement (GPM) daily precipitation product (0.1-degree grid resolution). By considering the temporal effect of rainfall, daily rainfall data were accumulated and overlaid using the GeoScene platform, and subsequently resampled to a 30-meter resolution. To facilitate high-speed surface computation and intuitive visualization in a matrix structure, the 14 indicators mentioned above were extracted and analyzed in raster format. As shown in Table 2, the sources and resolution of the 14 influencing factors are listed.

Methods

As illustrated in Fig. 5, the workflow of this paper is explicitly structured into three consecutive stages. Using pre- and post-rainfall high-resolution optical imagery, landslides were manually delineated through human–computer interactive interpretation. The resulting inventory was then assessed for completeness via area–number and area–frequency density relationships and analysed for spatial clustering patterns in terms of landslide point density and landslide area density. The validated inventory was coupled with fourteen rainfall-landslide conditioning factors (Elevation, Slope, Aspect, Topographic relief, Slope curvature, Profile curvature, Plane curvature, Strata, TWI, Distance to river, Land cover, Soil type, NDVI and Accumulated rainfall). Landslide points were extracted from the material source areas of landslides, and corresponding influencing factor data were retrieved. Four metrics—landslide area, landslide count, Landslide Number Density (LND) and Landslide Area Percentage (LAP)—were employed to quantify the distribution of landslides within individual factor classes, thereby revealing factor-specific susceptibility patterns. An equal number of non-landslide points were randomly sampled to balance the dataset. Pearson correlation analysis was first performed to examine multicollinearity among predictors. The AutoML framework (AutoGluon) was then applied to train 100 iterative models under two data-splitting schemes (70:30 and 80:20). Variable-importance rankings were extracted to identify the most influential factors, followed by two-factor and three-factor interaction analyses to elucidate synergistic amplification or non-linear suppression effects on landslide occurrence.

In this paper, rainfall-triggered landslide inventories were compiled through human–computer interactive interpretation, a technique widely recognized as mature in landslide data collection (Guzzetti et al. 2012; Wieczorek 1984;

Fig. 4 Fourteen influencing factors for rainfall-triggered landslides analyzed in this paper. **(a)** Elevation; **(b)** slope gradient; **(c)** aspect; **(d)** strata; **(e)** topographic relief; **(f)** normalized difference vegetation index (NDVI); **(g)** slope curvature; **(h)** plan curvature; **(i)** profile curvature; **(j)** topographic wetness index (TWI); **(k)** land cover; **(l)** soil type; **(m)** distance to rivers; **(n)** accumulated rainfall

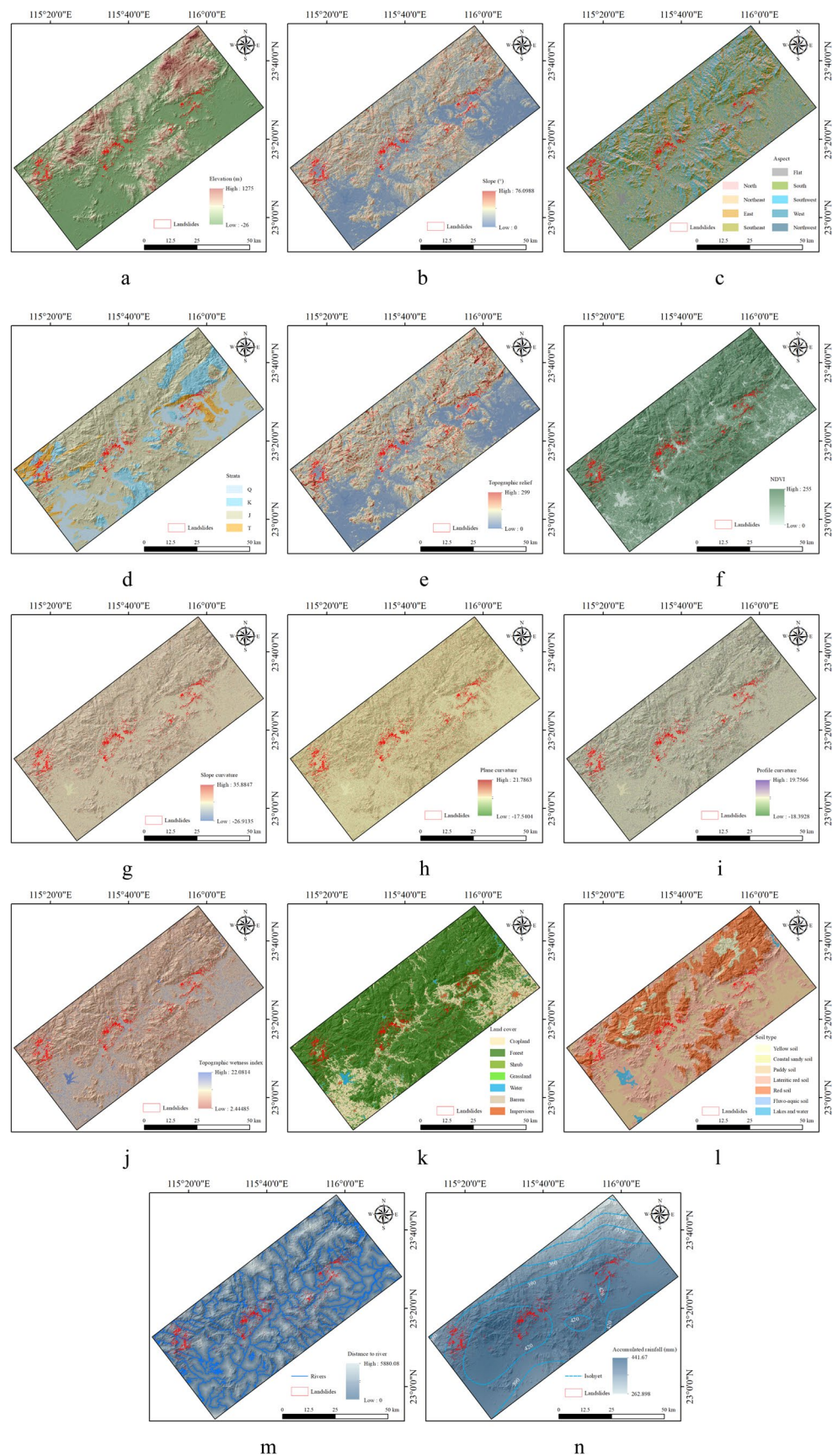


Table 1 Lithologic classification of the study area

No.	Strata	Lithologic classification
1	Quaternary(Q)	Silt, Sandy Silt, Fine Sand, Clay
2	Cretaceous(K)	Amphibolite, Gabbro, Diorite, Granite, Monzogranite, Granite Porphyry, Quartz Porphyry, Granodiorite, Granodiorite Porphyry, Conglomerate, Siltstone, Mudstone, Polymictic Conglomerate, Sandstone
3	Jurassic(J)	Monzogranite, Granite, Rhyolite Porphyry, Rhyolite, Basalt, Biotite Granite, Two-Mica Granite, Granite Porphyry, Granodiorite, Granodiorite Porphyry, Rhyodacite Porphyry, Dacite Porphyry, Carbonaceous Shale, Sandstone, Siltstone, Mudstone, Shale, Arkose, Silty Shale, Fine-Grained Sandstone
4	Triassic(T)	Conglomerate, Oil-Bearing Shale, Sandstone, Siltstone, Shale, Mudstone, Fine-Grained Sandstone, Silty Clay, Arkose

Table 2 Primary sources of the datasets

Data name	Data source	Spatial resolution
Elevation	https://search.asf.alaska.edu/	12.5 m
Slope	https://search.asf.alaska.edu/	12.5 m
Aspect	https://search.asf.alaska.edu/	12.5 m
Topographic relief	https://search.asf.alaska.edu/	12.5 m
Slope curvature	https://search.asf.alaska.edu/	12.5 m
Profile curvature	https://search.asf.alaska.edu/	12.5 m
Plane curvature	https://search.asf.alaska.edu/	12.5 m
TWI	https://search.asf.alaska.edu/	12.5 m
Strata	http://www.ngac.org.cn/	30 m
Distance to river	https://www.webmap.cn/	30 m
Land cover	https://zenodo.org/	30 m
Soil type	https://www.resdc.cn/	30 m
NDVI	https://www.nesdc.org.cn/	30 m
Accumulated rainfall	https://disc.gsfc.nasa.gov	30 m

Xu 2015; Xu et al. 2014b). The method exploits pre- and post-event high-resolution optical remote-sensing imagery to detect newly developed flow tracks, vegetation loss, and tonal contrasts that were absent prior to the rainfall event, with final delineation validated by expert analysts (Xie et al. 2025b; Xu et al. 2015).

Compared to traditional landslide assessment processes, which require manual feature engineering, parameter tuning, and model selection, as well as AutoML frameworks like AutoKeras and auto-sklearn that focus solely on unimodal tabular data (He et al. 2025; Ma et al. 2025), AutoGluon offers superior integrated performance through one-click multimodal end-to-end learning within the same development timeframe. Unlike physics-based models grounded in mechanical principles, purely data-driven methods like AutoGluon cannot explicitly incorporate geotechnical strength parameters, hydrological coupling processes, or three-dimensional topographic

stress fields (Qiu et al. 2025; Zhao et al. 2025). In this paper, model training and testing were conducted on the sample set using the AutoGluon automated machine-learning framework. AutoGluon's multi-layer stacking leverages a deep ensemble architecture, reuses original features through residual-like skip connections, and relies on rigorous out-of-fold (OOF) prediction mechanisms to exploit model diversity efficiently (Erickson et al. 2020). This design paradigm transcends traditional AutoML frameworks' dependence on Combined Algorithm Selection and Hyperparameter optimization (CASH), demonstrating that automated model ensemble integration itself constitutes a more effective pathway for accuracy enhancement. Figure 6 illustrates the core visualization of this strategy, depicting a two-tier stacking structure incorporating n base learners.

To comprehensively evaluate landslide occurrence characteristics across partitioned intervals, two widely adopted metrics were selected for developmental pattern analysis: Landslide Number Density (LND) and Landslide Area Percentage (LAP) (Ferrario 2022; Guzzetti et al. 2009; Xu et al. 2014a). LND quantifies spatial concentration by calculating the number of landslides per square kilometer (Eq. (1)). Conversely, LAP measures the areal impact within specific factor classes, defined as the percentage of landslide-affected area relative to the total area of each class (Eq. (2)). Statistical analyses were then employed to investigate relationships between rainfall-triggered landslide distribution and influencing factors.

$$LND = \frac{\text{Landslide Number}}{\text{Influence Factor Class Area}} \quad (1)$$

$$LAP = \frac{\text{Landslide Area}}{\text{Influence Factor Class Area}} \quad (2)$$

Results and analysis

Rainfall-induced landslide inventory

Based on human-computer interactive interpretation of pre- and post-event optical satellite imagery, a total of 6194 landslides were mapped within an area of 5012.41 km². Individual landslide areas range from 253.8 m² to 128394.8 m², yielding a cumulative landslide area of 12.69 km² and a mean landslide area of 2048.7 m². Point and area densities were subsequently estimated using a Gaussian kernel density estimator with a 5 km search radius. As shown in Fig. 7a and b, the maximum point density reaches 17.034 km⁻² and the maximum area density 3.898%, landslides are concentrated in the southwestern, central-south-eastern and north-eastern portions

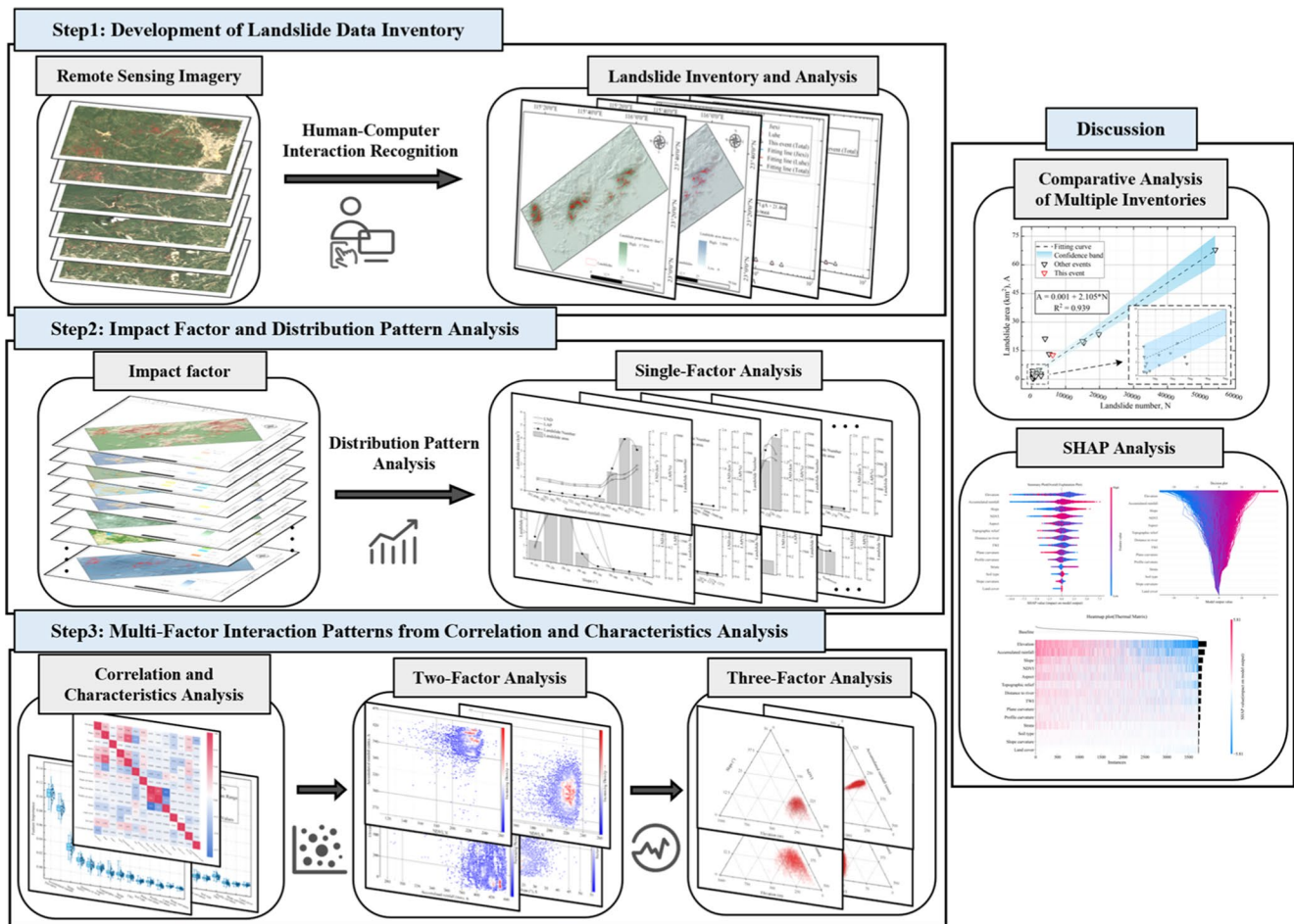


Fig. 5 Flowchart of the creation of the complete rainfall-induced landslide inventory and the analysis of multi-factor coupling patterns

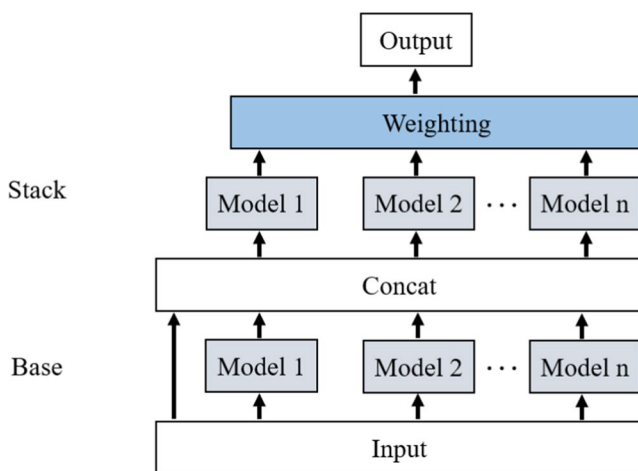


Fig. 6 Multi-layer stacking strategy of AutoGluon, illustrating a two-tiered architecture with n base learners

of the study area. These zones coincide with pronounced topographic relief and lie adjacent to the coast. Persistent onshore breezes transport humid air masses that,

upon orographic lifting, rapidly condense, resulting in elevated soil moisture in high-relief areas and thereby predisposing slopes to failure. Previous studies have examined landslide inventories for Jiexi County (Xie et al. 2023) and Luhe County (Li et al. 2024a). To enable direct comparison with these inventories, the present analysis adopts three metrics—landslide count, landslide area and frequency density—across Jiexi, Luhe and eastern Guangdong. Figure 7c displays the cumulative landslide count as a function of landslide area. The curve exhibits a gradual decline with increasing area, with the steepest decrease occurring in the 1000–2000 m² interval. Linear regression of this relationship for all three inventories yields correlation coefficients exceeding 0.96, and the landslides within this size range account for the majority of each inventory, indicating that the subset adequately represents overall landslide development. Figure 7d presents the frequency density versus landslide area on a log–log plot. The distribution follows a power-law decay after an initial rise, confirming the completeness of the compiled inventories.

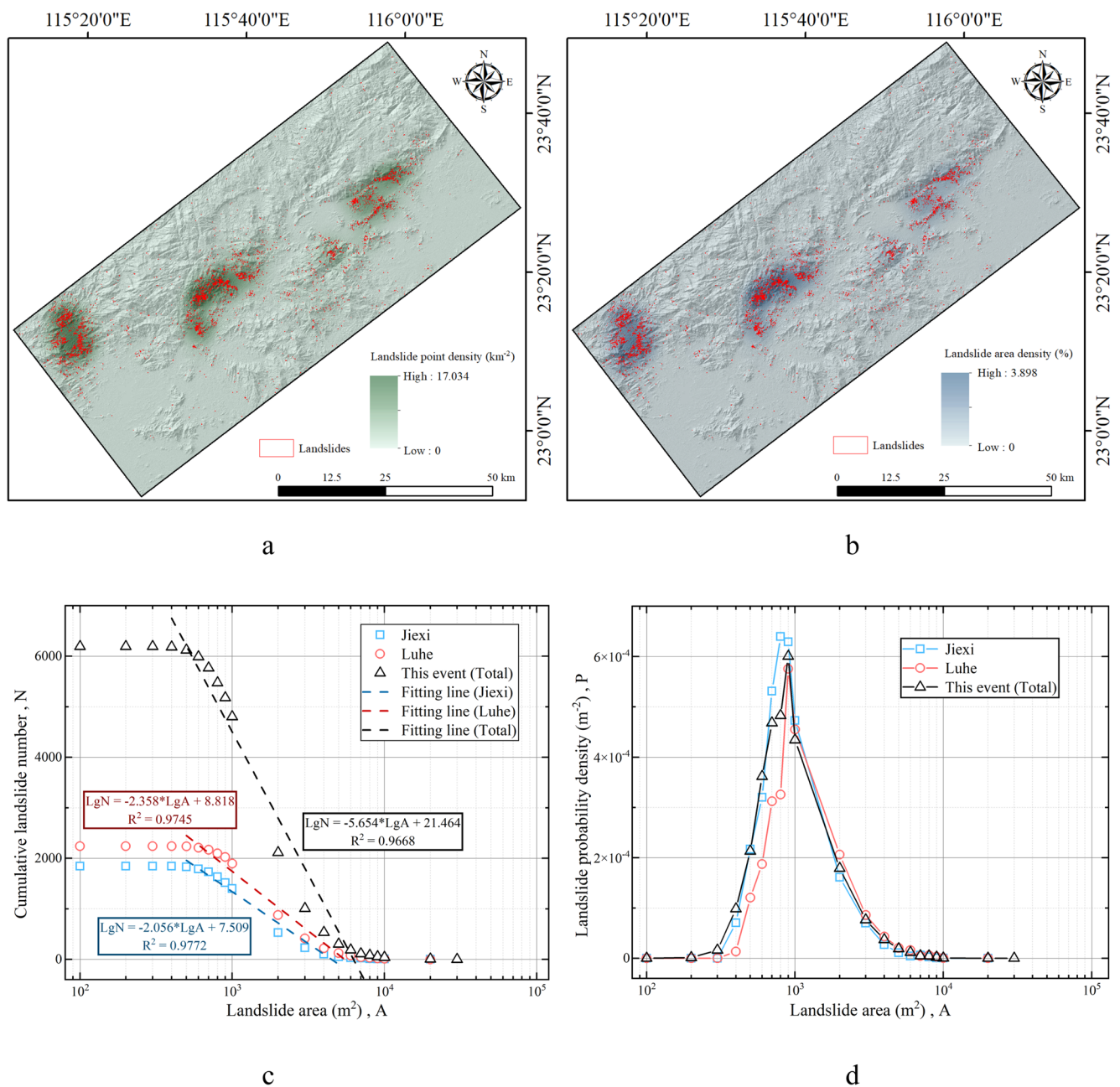


Fig. 7 Spatial analysis and size statistics of landslides in the study area. **(a)** Landslide point density; **(b)** landslide area density; **(c)** area-number distribution; **(d)** area-frequency density distribution

Spatial distribution pattern of landslides

Investigating landslide distribution patterns reveals relationships between landslides and various natural factors, elucidating their causal mechanisms and evolutionary processes. This paper analyzed landslide distribution patterns for 14 factors using four evaluation metrics: landslide area, landslide count, LND, and LAP. As shown in Fig. 8a, all four metrics exhibit a unimodal trend with increasing Elevation. Landslide area and count peak within 74–174 m, while LND

and LAP peak within 174–274 m. Landslides from the current event concentrate at 74–374 m, comprising 4884 landslides (78.85% of the total). Preliminary analysis indicates this elevation range likely combines steep slopes, intensive human activity, and efficient rainwater convergence, collectively enhancing landslide susceptibility. Figure 8b demonstrates that all metrics follow a similar unimodal trend with increasing Slope. Landslide area and count peak at 10–20°, whereas LND and LAP peak at 20–30°. Rainfall-triggered landslides concentrate within 10–30°, totaling

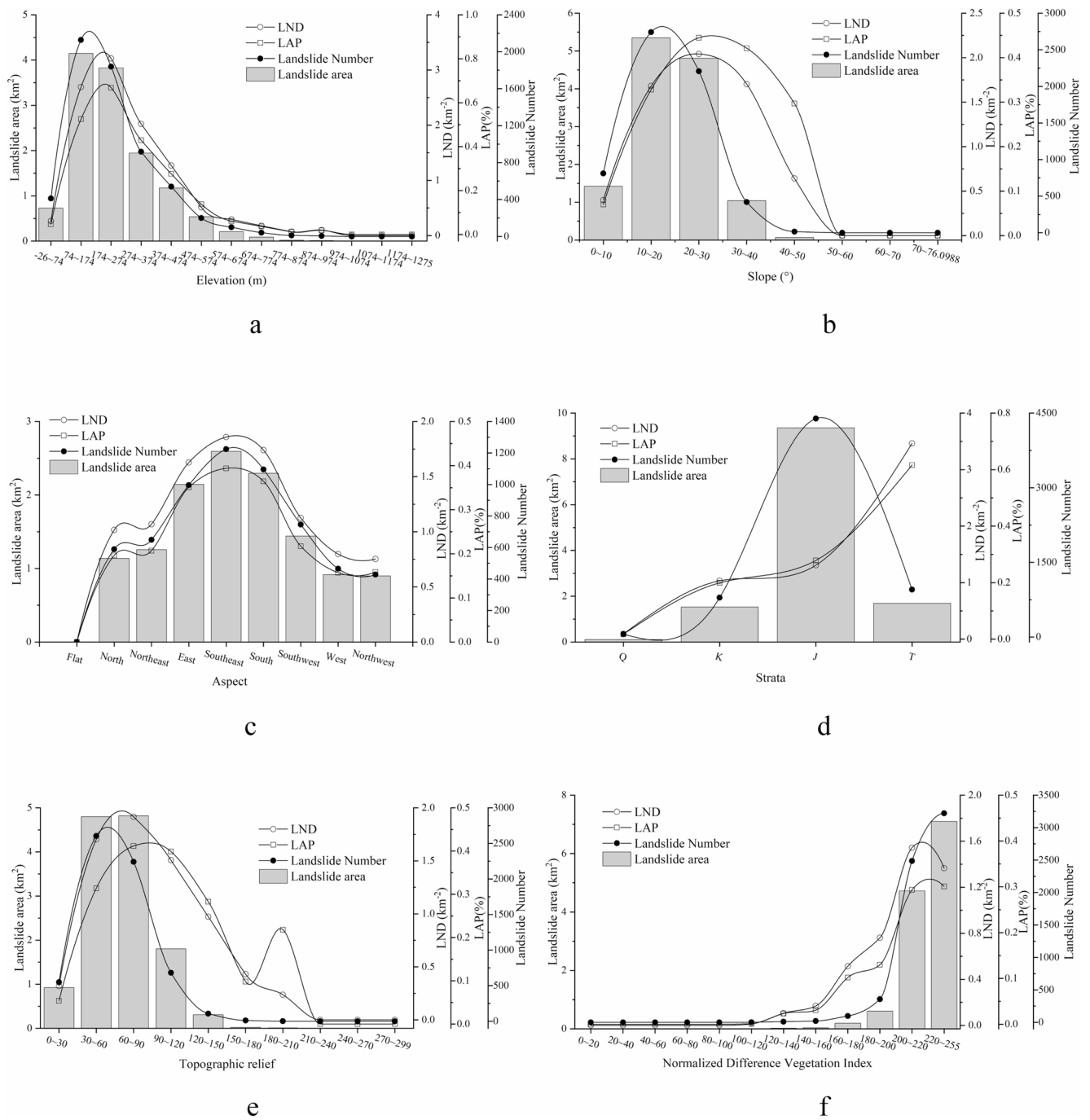


Fig. 8 The six influencing factors among the 14 factors used in this paper. (a) Elevation; (b) slope gradient; (c) aspect; (d) strata; (e) topographic relief; (f) normalized difference vegetation index (NDVI)

4948 landslides (79.88%). This distribution suggests, lower slopes experience groundwater convergence weakening soil strength, higher slopes may have achieved geomorphic stability through historical slope failures. Figure 8c reveals peak values across all metrics on Southeast-facing slopes (1225 landslides, 2.595 km², 1.86 km⁻², 0.394%), followed by South and East aspects. Due to the lower terrain

in the southeast and higher terrain in the northwest, with the southeastern area being coastal, the southeast-facing slopes in the region become windward slopes. Windward slopes are located in the area where air currents are forced upwards, making them prone to precipitation accumulation. In contrast, leeward slopes experience less precipitation due to the downward movement of air. During rainfall

events, windward slopes typically receive more precipitation, leading to an increase in soil moisture saturation in the region, which in turn promotes a rise in pore water pressure. When pore water pressure increases, the soil is more likely to undergo shear failure, thus increasing the probability of landslide occurrence. As shown in Fig. 8d, within the Strata classification, landslide count and area peak in the J interval (4391 landslides, 9.355 km²), while LND and LAP reach their maxima in the T interval (3.464 km⁻², 0.616%). Landslide concentration in the T interval significantly exceeds other strata due to its characteristic of interbedded layers with varying thicknesses, which reduce shear strength when saturated. The J interval contains the largest number of landslides (4391, 70.89% of total), primarily developing in granite formations featuring low geomechanical strength, poor stability, and permeable weathered soil layers. Rainfall readily infiltrates but drains poorly through these materials, progressively destabilizing the rock-soil mass. Figure 8e demonstrates that all four metrics follow a unimodal trend with increasing Topographic relief. Landslide count peaks within 30–60 units, while the other three metrics peak within 60–90 units. Landslides concentrate at 30–90 units (4846 landslides, 78.24%). This distribution suggests, Low-relief terrain facilitates rainwater accumulation, infiltration, and retention, substantially weakening soil strength. High-relief terrain typically exhibits exposed bedrock or thin overburden, where competent rock provides high stability and efficient drainage minimizes water accumulation, resulting in stable geomorphic features. Figure 8f reveals that all metrics increase with higher NDVI values. LND peaks within 200–220, while the other metrics peak within 220–255. Landslides concentrate at NDVI 200–255 (5709 landslides, 92.17%). Although vegetation is generally regarded as a stabilizing factor for slopes due to its root reinforcement and transpiration, under specific conditions, its own characteristics can become a major cause of instability. First, the root network, while enhancing the soil's shear strength, also creates preferential flow paths that facilitate the rapid infiltration of precipitation, thereby sharply increasing pore water pressure at the potential sliding surface. Secondly, the load of vegetation and the dynamic forces transmitted by wind-induced effects can exert additional static and dynamic moments on the soil, potentially exacerbating the slope's instability. This pattern is attributed to well-developed root systems in high-NDVI areas accelerating deep soil infiltration during rainfall. This process significantly elevates pore-water pressure and reduces shear strength along potential failure surfaces, thereby increasing landslide susceptibility.

As illustrated in Fig. 9a and c, all four evaluation indices exhibit an initial rise followed by a decline with increasing slope curvature, plan curvature and profile curvature. Landslide counts and areas peak within the ranges of –2–4

and –1–3 respectively, whereas LND and LAP reach maxima at –8– –2 and –5– –1. Within the profile curvature domain, landslide count, area and LAP peak between –2 and 2, while LND peaks between 2 and 6. Statistically, the event's landslides therefore concentrate within a relatively narrow curvature band, low values correspond to gentle terrain unfavorable to failure, whereas high values reflect steep topography where natural shedding has already occurred, leaving only the intermediate curvature range where rainfall acts as the necessary trigger. Figure 9d further demonstrates a unimodal response to TWI, all indices peak between 4 and 6 (4232 landslides, 8.392 km², 1.538 km⁻² and 0.305%). Interpretively, zones of high TWI retain abundant soil moisture and have already undergone prior sliding or adjustment, attaining relative stability. Conversely, in zones of low to moderate TWI, rapid water infiltration during rainfall sharply elevates saturation, precipitating failure.

Figure 10a shows that within the forest land-cover class the indices attain their maxima (6042 landslides, 12.4 km², 1.563 km⁻² and 0.321%). The concentration of rainfall-triggered landslides in forested areas is attributed to dense root networks that facilitate preferential infiltration, increase pore-water pressure along potential sliding surfaces and accelerate soil softening and disintegration, thereby promoting failure. As shown in Fig. 10b, within the lateritic red soil range of the Soil type category, the number of landslides, landslide area, LND, and LAP all reached their peaks. The specific values were 4121 landslides, 8.31 km², 1.968 km⁻², and 0.397%, respectively. Landslides triggered by this rainfall event were predominantly concentrated within the lateritic red soil range. In South China, lateritic red soil exhibits high clay content, fine particles, large specific surface area, and strong adsorption capacity. These properties make it prone to water absorption, swelling, and softening, which reduce shear strength and diminish inter-particle friction and cohesion, thereby increasing landslide risk (Huang et al. 2022). Furthermore, lateritic red soil is susceptible to developing desiccation cracks when dry, disrupting soil continuity (Zhang et al. 2025b). These cracks provide pathways for rapid rainwater infiltration, causing a sharp increase in soil moisture content during rainfall and further triggering landslides. As illustrated in Fig. 10c, both the number of landslides and landslide area initially increased and then decreased with increasing distance to river. In contrast, LND and LAP exhibited two fluctuations as the distance increased. Specifically, the number of landslides peaked within the 500–1000 m range. Landslide area reached its maximum within the 1000–1500 m range. LND and LAP peaked within the 1500–2000 m range. Landslides triggered by this rainfall were mainly concentrated within 0–2000 m from rivers, totaling 5474 landslides (88.38% of the total). Preliminary analysis suggests that soil in areas

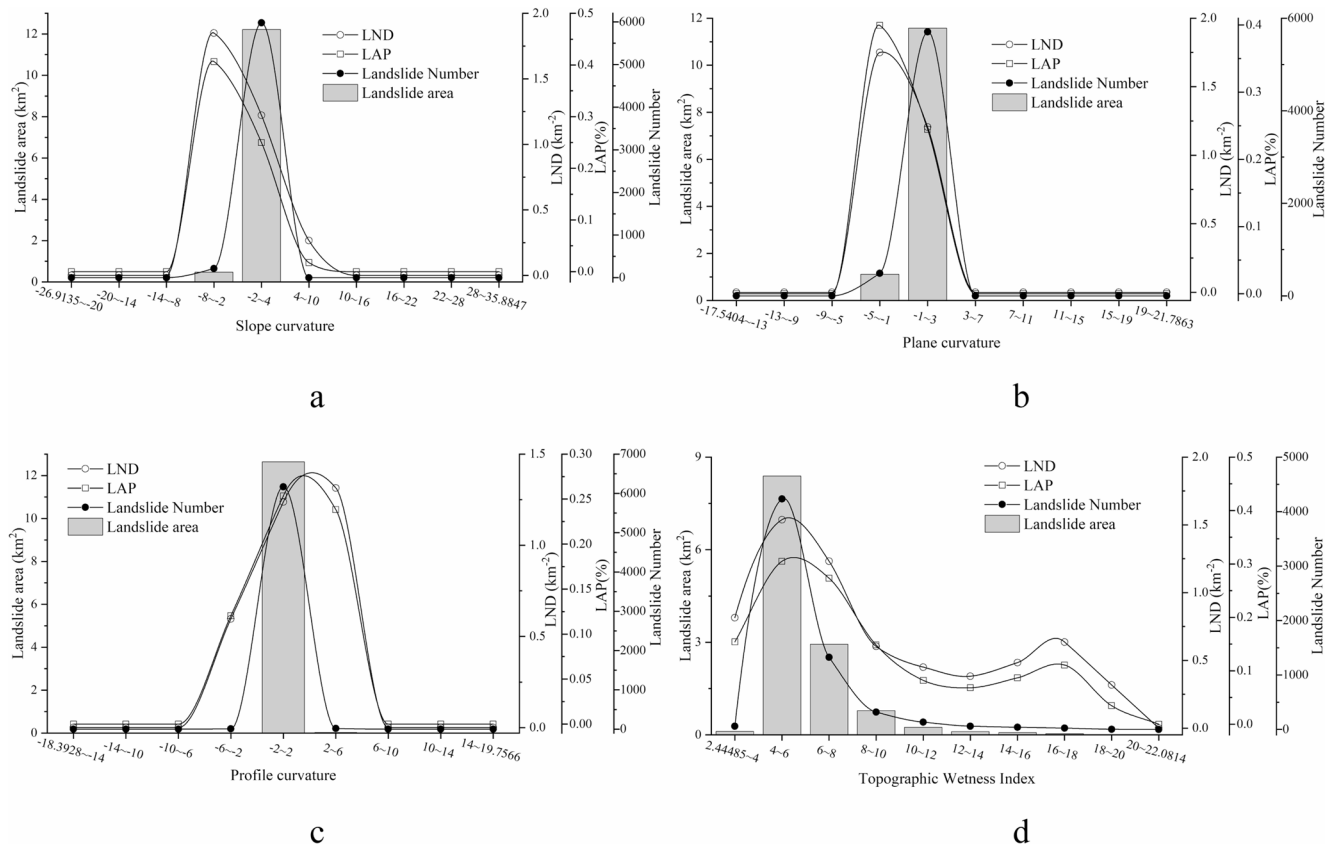


Fig. 9 The four influencing factors among the 14 factors used in this paper. (a) Slope curvature; (b) plan curvature; (c) profile curvature; (d) topographic wetness index (TWI)

close to rivers is prone to erosion by water flow, leading to loosened soil structure, reduced stability, and consequently increased landslide susceptibility. As depicted in Fig. 10d, both the number of landslides and landslide area peaked within the 402–422 mm rainfall accumulation range, with values of 2539 landslides and 5.131 km², respectively. The maximum LND (2.216 km⁻²) and LAP (0.482%) occurred within the 422–441.67 mm range. Landslides induced by this event were overwhelmingly concentrated within the 382–441.67 mm rainfall accumulation range, accounting for 6007 landslides (96.98% of the total). Preliminary analysis indicates that higher cumulative rainfall increases soil moisture content and decreases soil shear strength, making landslide occurrence more likely. Additionally, landslides in this event may have been influenced by other factors such as topography, geological setting, and human activities.

Factor correlations and weighting

To enable subsequent multi-factor coupling analysis, machine learning models require training on both landslide and non-landslide samples. AutoGluon accommodates this through automated processing of heterogeneous data,

multi-layer stacking ensembles, and robust fault tolerance mechanisms. This framework maintains high accuracy while ensuring computational resilience, effectively capturing the complex nonlinear relationships inherent in landslide data. As shown in Fig. 11, our study utilized 6,194 landslide samples triggered by the rainfall event. Negative samples were constructed by randomly selecting 6,194 non-landslide points (1:1 ratio) outside landslide-affected zones.

This paper employed Pearson correlation analysis to assess inter-factor relationships across the 14 influencing factors characterizing both positive (landslide) and negative (non-landslide) samples. As shown in Fig. 12a, the strongest positive correlation (0.87) occurred between slope curvature and plan curvature, followed by topographic relief and slope gradient (0.83). The most significant negative correlation (−0.87) was observed between slope curvature and profile curvature. All other pairwise Pearson coefficients fell within moderate ranges. This analysis aimed to eliminate highly correlated variables, thereby preventing multicollinearity effects during subsequent multi-factor coupling investigations. The integrated dataset (landslide and non-landslide samples) was processed using the AutoGluon framework for model training and testing.

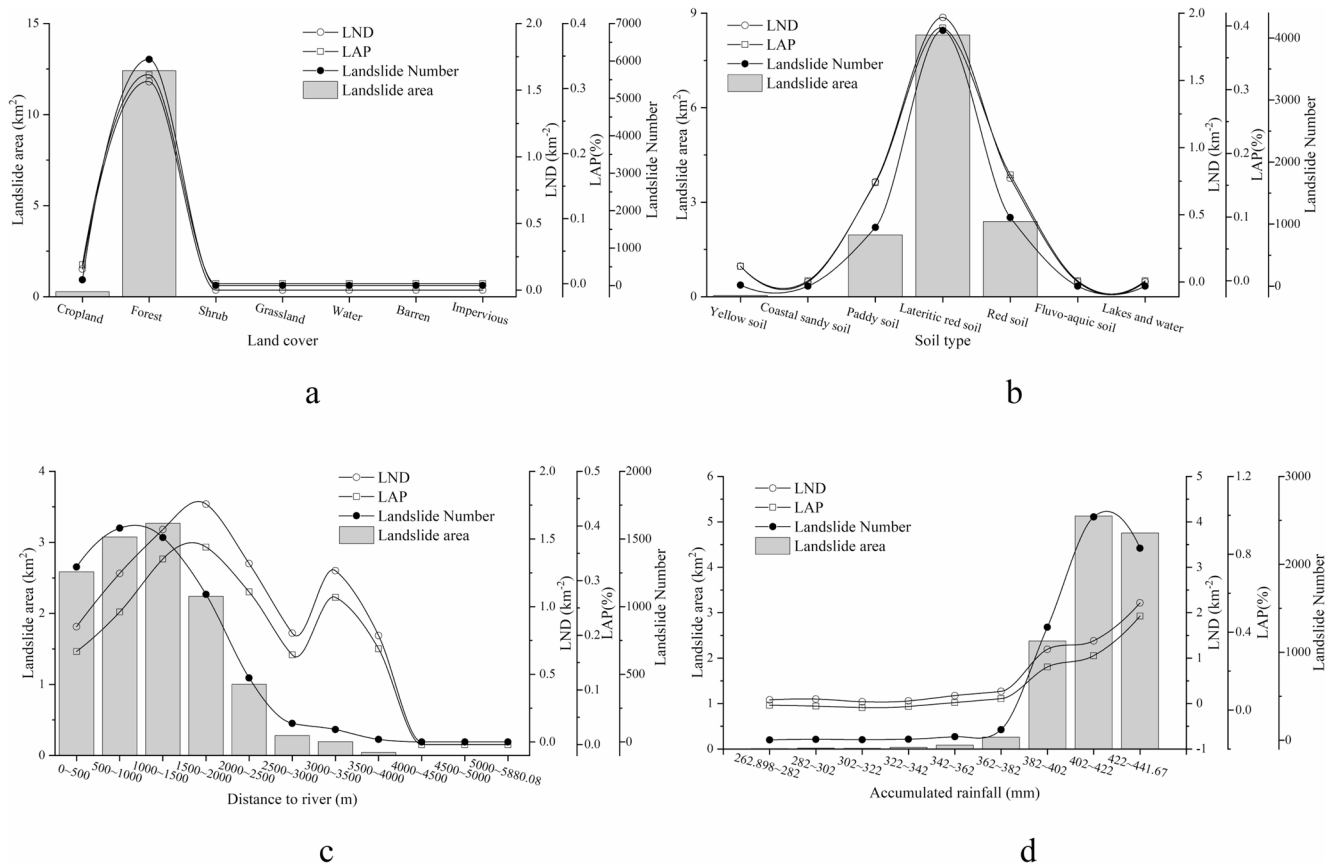


Fig. 10 The four influencing factors among the 14 factors used in this paper. **(a)** Land cover classification; **(b)** soil type; **(c)** distance to rivers; **(d)** accumulated rainfall

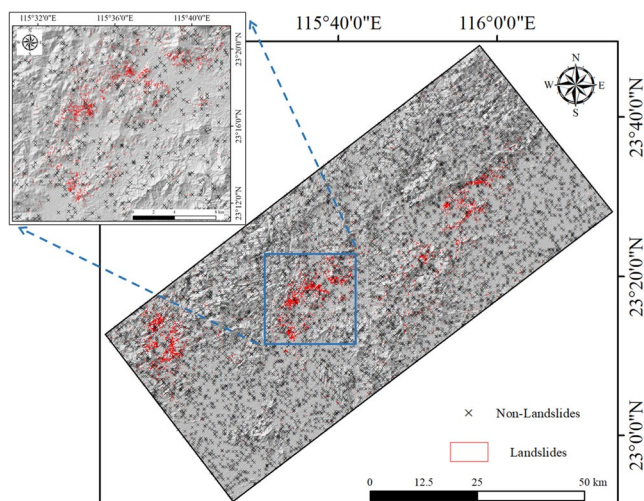


Fig. 11 Landslide distribution map and spatially balanced non-landslide samples (1:1 ratio)

The AutoGluon TabularPredictor was used with the mode set to “best_quality” and a time budget of 60 min (early_stop_patience=10). Core parameters were automatically iterated within the search space by the framework. The

base learners included six families: LightGBM, CatBoost, XGBoost, RandomForest, ExtraTrees, and NN_Torch. Each family’s hyperparameter combinations were optimized through approximately 50 to 80 trials using Bayesian optimization until the validation set AUC showed no improvement. During the stacking phase, 10-fold bagging and 5-layer stacking were enabled by default, repeated 3 times (full_refit=True) to reduce variance. Class imbalance was automatically handled by AutoGluon. Adaptive sampling and threshold shifting were applied to the samples, eliminating the need for manual weighting. To prevent information leakage, all features were first split using stratified sampling, with only the training set involved in the AutoGluon fitting process. The test set was input only during the final evaluation. The entire process was fixed with random_seed=42, and the execution environment was Windows 10/AMD A12-9700P/16GB RAM, with Python 3.9. Figure 12b presents ROC curves and corresponding AUC values for AutoGluon’s 14 constituent models. The WeightedEnsemble model achieved the highest AUC (0.8789), closely followed by CatBoost (0.8775). These robust model performances satisfy the prerequisites for feature importance analysis in subsequent research phases.

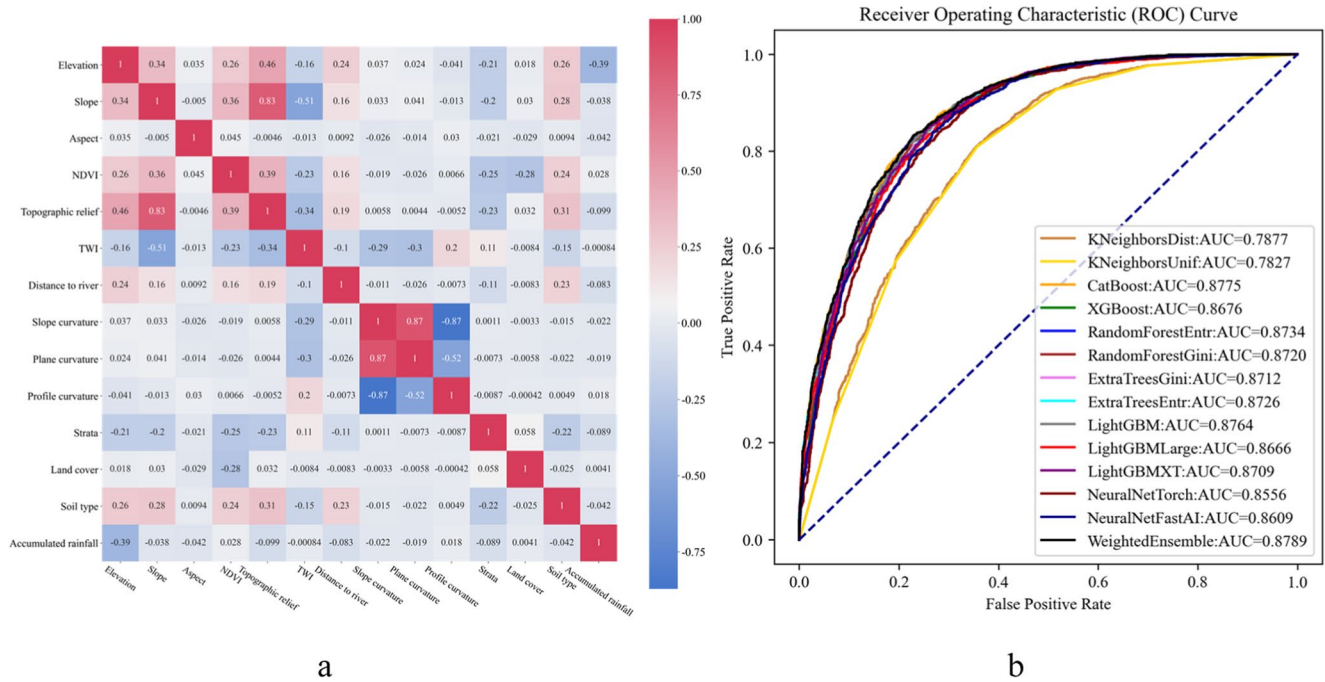


Fig. 12 Correlation analysis of the sample set and ROC curves with AUC values. **(a)** Pearson correlation analysis for training and testing sets; **(b)** ROC curves with AUC values for the testing set

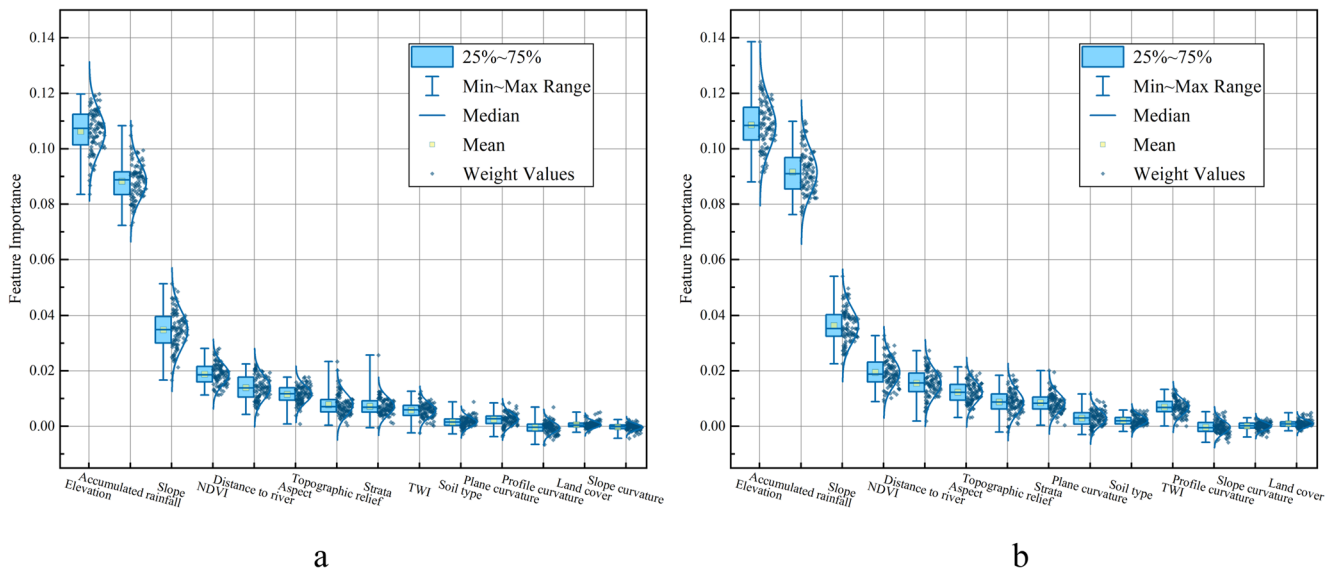


Fig. 13 Weighted feature importance rankings from 100 iterations of AutoGluon's WeightedEnsemble model under different training-testing partitions. **(a)** 70:30 partition ratio; **(b)** 80:20 partition ratio

In this section, landslide and non-landslide samples were randomly split into 100 training–testing pairs at a 7:3 ratio. Using the WeightedEnsemble model—the best-performing algorithm in the AutoGluon framework—100 sets of factor-importance weights were obtained. As shown in Fig. 13a, Elevation consistently exhibited the highest median weight (0.10737) and mean weight (0.10626), with

the majority of values falling between 0.1014 and 0.11246. Accumulated rainfall ranked second, with a median of 0.08881 and a mean of 0.08829, and most weights ranging from 0.08348 to 0.09166. The same procedure was repeated after re-partitioning the samples into 100 8:2 training–testing pairs. Figure 13b indicates that Elevation again dominated (median=0.10839, mean=0.10856,

inter-quartile range=0.10315–0.11493), followed by Accumulated rainfall (median=0.09096, mean=0.09161, inter-quartile range=0.08551–0.09681). Comparing the distributions of weights generated under the two prevailing data-splitting schemes allows an objective assessment of the robustness of factor-importance rankings across random partitions. Overall, the four most influential environmental factors controlling landslide initiation in this paper are Elevation, Accumulated rainfall, Slope, and Normalized Difference Vegetation Index.

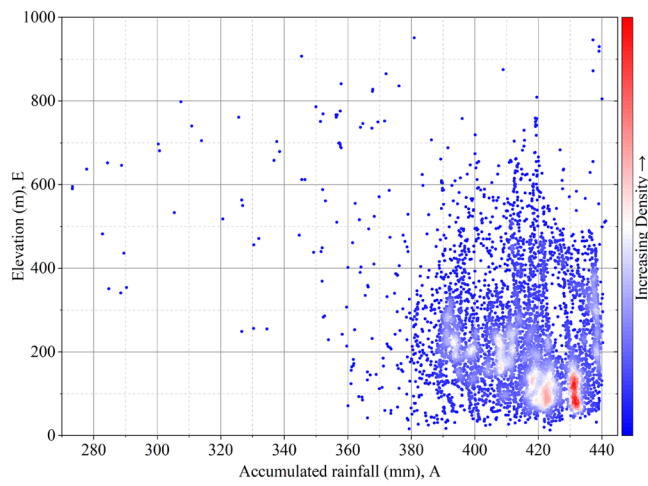
Multifactor interaction mechanisms

To deepen understanding of landslide distribution patterns, the research statistically examined the density response of landslides to all pairwise combinations of Elevation, Accumulated rainfall, Slope and NDVI—the four dominant factors ranked by AutoGluon-derived importance. The highest landslide counts concentrate in the bin of 20–400 m elevation combined with 380–440 mm accumulated rainfall (Fig. 14a), because low-to-mid elevations accumulate colluvium and are subject to intensified human disturbance, while orographic lifting funnels additional rainfall downslope, rapidly raising pore-water pressure. The same elevation range paired with slopes of 10°–25° also yields the greatest density (Fig. 14b), as weathered regolith is abundant and the moderate gradient situates slopes near the critical equilibrium state, further destabilised by enhanced precipitation. When the 20–400 m elevation band intersects NDVI values of 210–230, landslide counts again peak (Fig. 14c), owing to dense root networks that create preferential infiltration paths to deep-seated slip surfaces and to locally intensified rainfall that quickly saturates soils, increasing both weight and driving forces. Similarly, the combination of 380–440 mm rainfall with NDVI 210–230 produces the highest landslide frequency (Fig. 14d), because vegetation roots rapidly convey water beyond canopy interception along macropores, saturating weathered zones and elevating pore-water pressure. The interaction of 380–440 mm rainfall with slopes of 10°–25° shows the greatest landslide concentration (Fig. 14e). Near-critical slope angles allow rapid infiltration to reduce effective stress while additional water mass increases gravitational loading, jointly promoting failure. Finally, moderate slopes (10°–25°) combined with NDVI 210–230 exhibit the largest landslide density (Fig. 14f), as the slope angle places the terrain at the threshold of stability, and dense root channels act as preferential conduits that accelerate infiltration and concentrate pore-water pressure along potential failure planes.

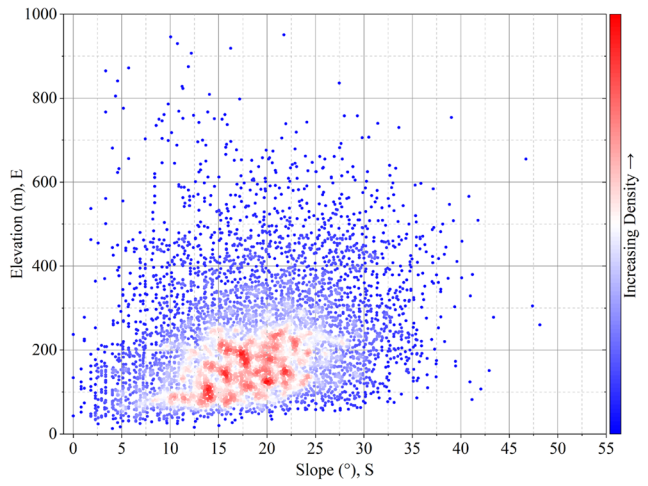
The occurrence of landslides results from interactions among multiple environmental factors. In low-to-medium elevation zones, the “orographic rainfall amplification

effect” induced by topographic uplift generates higher precipitation compared to surrounding areas, facilitating soil saturation. Simultaneously, moderate slope angles position hillslopes near critical static equilibrium thresholds, elevating failure susceptibility. Furthermore, while dense vegetation provides surficial consolidation through root networks, its macropore system creates preferential infiltration pathways that accelerate pore-water pressure buildup along potential failure planes. Consequently, regions characterized by low-to-medium elevation, moderate slopes, dense vegetation, and intense rainfall constitute high-density landslide hotspots. A landslide group in Tianshui, Gansu, is located in a low-to-middle elevation range (Zhou et al. 2025). During the season of concentrated rainfall and high cumulative precipitation, when the slope is within a moderate range, some landslide bodies exhibit significant displacement. The frequency of landslide occurrence in this range is much higher than in high-elevation or low-precipitation areas. Monitoring data indicate a substantial increase in pore water pressure in the loess layer, accompanied by a noticeable decrease in soil shear strength, confirming the landslide accumulation effect under the combined conditions of low-to-middle elevation, heavy rainfall, and moderate slopes.

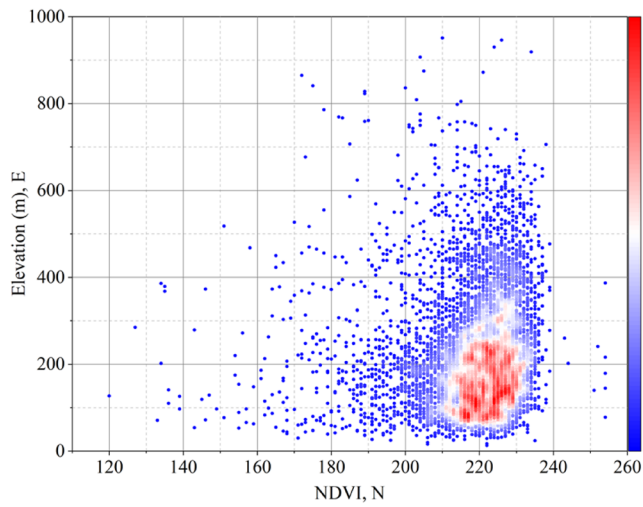
This paper continued by selecting the top four environmental factors ranked by the aforementioned weight values and analyzed the contribution of different ternary factor combinations to landslide development. Through comparative analysis and mutual comparison of the binary factor results, the coupling relationships between factors were clearly elucidated. As shown in Fig. 15a, areas with low-to-mid elevation, moderate slopes, and intense rainfall are zones of high landslide density. This finding is consistent with the conclusions drawn from the binary factor analysis. Similarly, Fig. 15b shows that areas with low-to-mid elevation, moderate slopes, and dense vegetation also exhibit high landslide density, aligning with the previous conclusions derived from binary factor analysis. Figure 15c reveals that regions characterized by low-to-mid elevation, vegetation cover, and moderate rainfall are also prone to high landslide density. While the elevation condition for dense landslide distribution matches the previous results, the NDVI and accumulated rainfall values fall outside their typical ranges and exhibit limited promoting effects. This outcome primarily indicates that elevation is the dominant controlling factor for landslide occurrence, which, besides the three existing environmental factors, is also influenced by other elements. Furthermore, comparing Fig. 15a and c reveals that when controlling for slope and NDVI, the landslide density distribution shows dispersion in one case and convergence in the other. This reflects that, in this landslide event, the contribution of slope to landslides is greater than that of



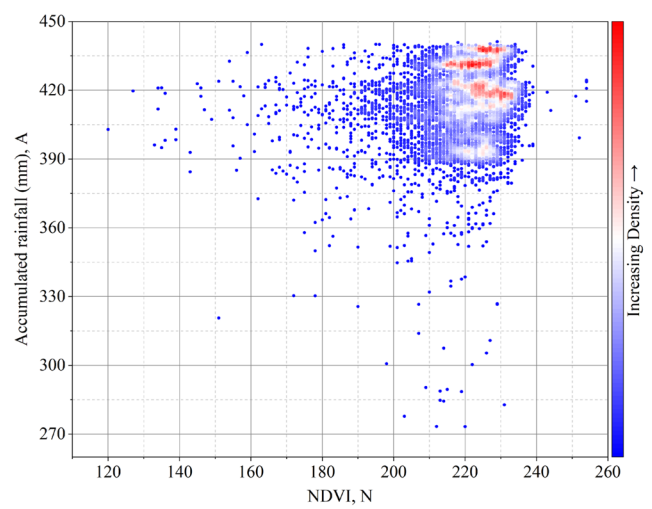
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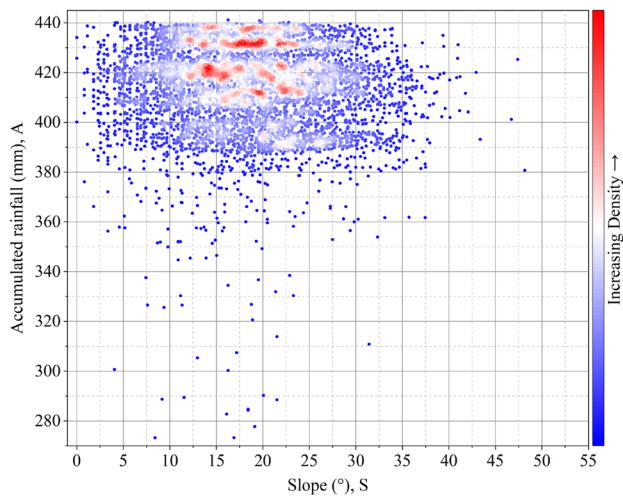
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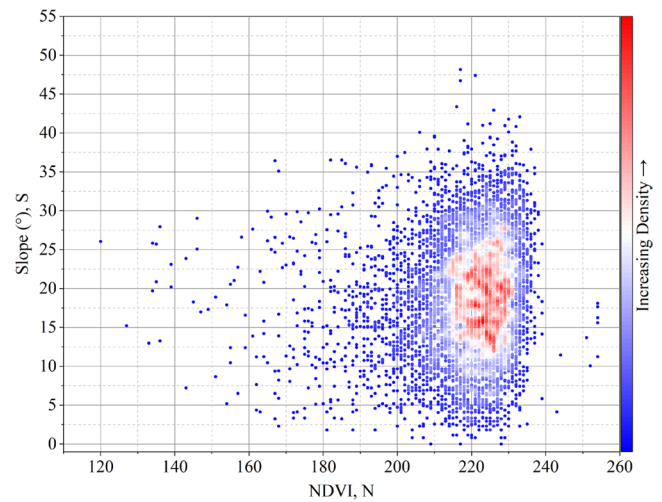
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f

Fig. 14 Spatial density distribution and trend analysis of landslides under different bivariate factor combinations. (a) Elevation–accumulated rainfall; (b) elevation–slope gradient; (c) elevation–NDVI; (d) accumulated rainfall–NDVI; (e) accumulated rainfall–slope gradient; (f) slope gradient–NDVI

NDVI. As illustrated in Fig. 15d, areas featuring moderate slopes, vegetation cover, and moderate rainfall also correspond to high landslide density. The slope condition for dense landslides aligns with previous results. However, the NDVI and accumulated rainfall values again lie outside their typical ranges, showing negligible promoting effects. The absence of the dominant elevation factor results in suboptimal distribution criteria. Slope, as a representative of topographic variation, can partially reflect changes in elevation and thus falls within a suitable range in this case. Combining the observations from Fig. 15c and d, both accumulated rainfall and NDVI fall outside their optimal ranges for landslide development, and their distribution areas are mutually exclusive. Conventionally, landslides are less likely to occur in densely vegetated areas but more likely under heavy rainfall. The observed phenomenon—where dense vegetation coincides with landslide occurrence—is attributed to root channels accelerating rainwater infiltration, thereby increasing pore water pressure at potential failure surfaces and triggering landslides.

As shown in Fig. 15a, when considering only topography and rainfall, the landslide density distribution still aligns with the conditions established in previous conclusions. This indicates that the gravitational potential energy imparted to slopes by topography is the primary triggering factor. Rainfall increases pore water pressure, thereby exceeding the critical point of static equilibrium and elevating landslide risk. Figure 15b demonstrates that when considering only topography and vegetation cover, the landslide density distribution also conforms to previously established conditions. This suggests that slopes were already predisposed to landslide development prior to the influence of rainfall as a triggering factor. Among the combinations examined, both ‘low-to-medium elevation–gentle slope–intense rainfall’ and ‘low-to-medium elevation–gentle slope–high vegetation coverage’ are more prone to trigger landslides. A landslide along the highway in Hunan lies within a low-to-middle elevation range corresponding to relative elevation (Meng et al. 2022). When the cumulative rainfall reaches a high level and the vegetation coverage index is within a high-value range, significant displacement is observed both on the surface and in the deep layers of some landslide bodies. Analysis through X-ray fluorescence spectroscopy and X-ray diffraction tests revealed that the root systems in areas with high vegetation coverage accelerate rainfall

infiltration, causing a significant decrease in the strength of weak interlayers containing specific mineral components. This fully demonstrates the disaster-causing effect of the coupling between heavy rainfall and high vegetation coverage.

Discussions

Landslide inventory comparative analysis

To quantitatively assess the scale of regional clustered landslides triggered by intense rainfall, this paper systematically collected detailed data from 20 global rainfall-triggered clustered landslide events. These cases cover eight provincial-level regions in China (Anhui, Beijing, Fujian, Gansu, Guangdong, Hunan, Zhejiang, Chongqing) and internationally representative areas, including Italy and Brazil. Through standardized processing of disaster information, a statistical relationship between landslide count (N) and total landslide area (A) was established, as shown in Fig. 16. Data from 19 rainfall-triggered clustered landslide events (black triangles), excluding the current event (red triangle), were fitted. The results reveal a strong linear relationship among these data points. The correlation coefficient of the fitted line reached 0.939, demonstrating a robust correlation between landslide count and landslide area and indicating a satisfactory fit. Conversely, the data point for the current landslide event deviates significantly from the fitted curve. Preliminary analysis suggests this deviation may be attributed to the exceptionally prolonged rainfall duration in this case. This likely enhanced landslide mobility, resulting in extended runout distances and consequently increasing the total landslide area. It is noteworthy that the study area is located in South China, where short-term extreme rainfall events are frequent in the summer. Compared to data from 19 other rainfall-induced landslide cases, the slope of this event is steeper, which also suggests that the average landslide area caused by this event is larger than those of the other cases. This difference is likely due to the distinct triggering mechanisms of the events. Short-term extreme rainfall, with its extremely high rainfall intensity, promotes the rapid infiltration of rainwater into deeper layers through preferential flow paths, bypassing relatively impermeable layers. This leads to a sharp increase in excess pore water pressure on the potential widespread sliding surface, triggering large-scale, cohesive deep-seated sliding. In contrast, rainfall with a shorter duration has limited infiltration depth, primarily saturated shallow soil layers and causing the loss of matric suction, leading to localized, small-scale shallow instability. To mitigate discrepancies observed across

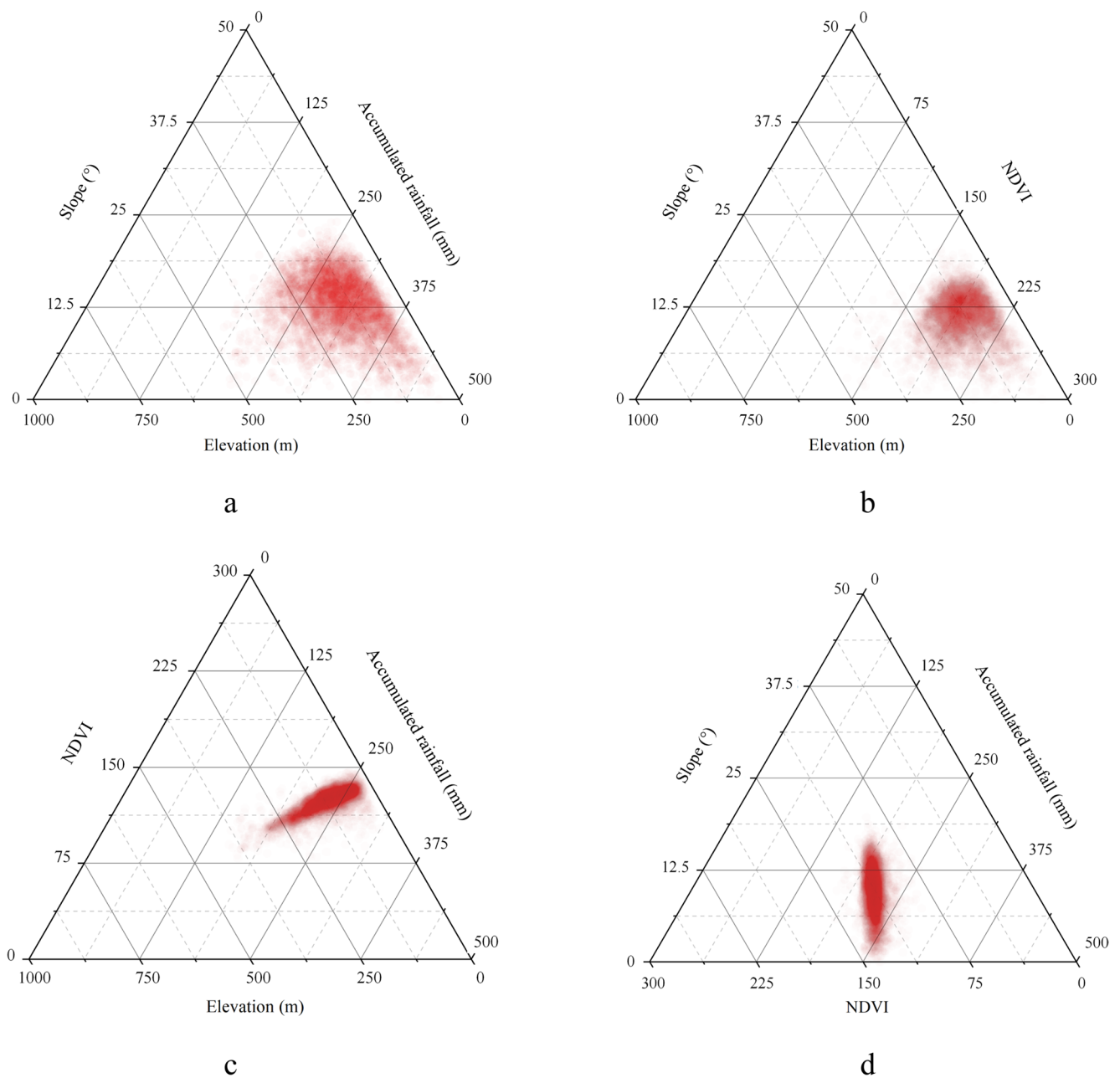


Fig. 15 Analysis of the contribution of different ternary factor combinations to landslide development. **(a)** Slope–accumulated rainfall–elevation; **(b)** slope–NDVI–elevation; **(c)** NDVI–accumulated rainfall–elevation; **(d)** slope–accumulated rainfall–NDVI

different inventories and enhance the reliability and precision of future landslide assessments, the following approaches are essential. Developing self-learning mechanisms for assessment models to automatically adjust parameters using new data, ensuring timeliness and accuracy (Bai et al. 2022; Liang et al. 2025; Zeng et al. 2025). Constructing real-time data-driven landslide risk assessment models to dynamically capture evolving risks (Liu et al. 2025b; Manconi and Giordan 2016; Rawat and Barthwal 2024). Leveraging large-scale AI model capabilities

to develop integrated intelligent disaster prevention and mitigation assistance systems for rapid response and decision support (Xie et al. 2025a; Yu et al. 2024).

SHAP analysis

With advancements in machine learning interpretability, SHapley Additive exPlanations (SHAP) has become widely adopted for evaluating the importance and influence direction of environmental factors in landslides (Pradhan

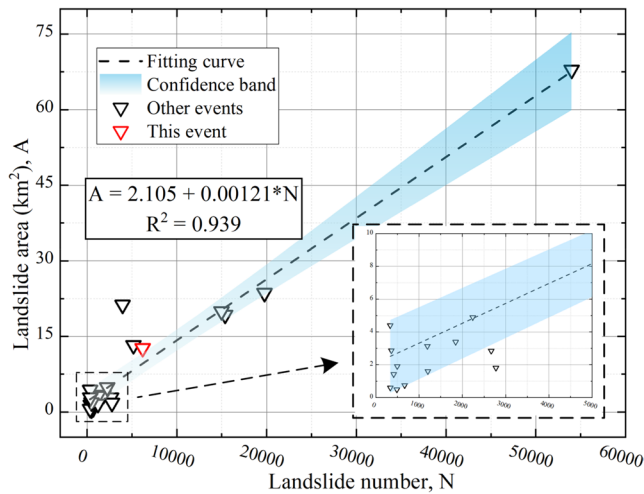


Fig. 16 Comparative analysis of area and count for rainfall-triggered landslide inventories across different regions

et al. 2023; Zhang et al. 2023). This paper employed the XGBoost model within the Autogluon framework to conduct SHAP analysis on the test sample set. As shown in the three subplots of Fig. 17, the top four factors ranked by importance are Elevation, Accumulated rainfall, Slope, and NDVI. Figure 17a illustrates that each point represents a landslide sample, where redder colors indicate higher factor values and bluer colors denote lower values. For Elevation's contribution, points exhibiting weak red with predominantly blue hues correspond to high SHAP values. For Accumulated rainfall, red points align with high contribution levels. Figure 17b visualizes the decision process for test set predictions. Each influencing factor assigns an estimated value to the prediction, and the superposition of these values determines the final output prediction. Notably, when the decision line traverses the intervals of Elevation and Accumulated rainfall, it exhibits distinct acceleration, indicating these two factors substantially contribute to the prediction outcomes. Figure 17c demonstrates the cumulative process of heat values for each sample. Factors with higher contributions display brighter heat values and larger SHAP values, whereas those with lower contributions appear dimmer with smaller SHAP values. Elevation, Accumulated rainfall, and slope all exhibit bimodal red-blue divergence, signifying these three factors play a decisive role in sample contributions and confirming their high importance. The annual maximum NDVI used in this paper may not fully represent the real-time vegetation conditions during specific events. This mismatch in time scale could be one of the reasons why NDVI did not perform as expected in the model. However, given the availability, data quality, and representativeness of NDVI data, the source used in this paper is currently the best available.

Verification of machine learning results

In the analysis of landslides triggered by this rainfall event, we employed an automated ensemble machine learning approach and considered 14 common influencing factors. Through the analysis, we identified five key factors—elevation, accumulated rainfall, slope, NDVI, and aspect—that play significant roles in landslide assessment, as evidenced by their consistent impact across landslide studies in different regions (Xie et al. 2025c). Firstly, elevation, as an important influencing factor, has received widespread attention in landslide assessments. Research shows that elevation significantly affects landslide occurrence and plays a crucial role in landslide prediction models, making it an indispensable factor in landslide evaluation (Addis 2023; Chen et al. 2018; Long et al. 2021). Following this, accumulated rainfall is considered a key factor in landslide assessment. Numerous studies have demonstrated a close relationship between heavy rainfall, accumulated rainfall, and landslide occurrence (Ng et al. 2021). An increase in the duration and total amount of rainfall typically reduces slope stability, thereby increasing the likelihood of landslides (Lizárraga and Buscarnera 2020). The importance of rainfall in landslide assessments has been repeatedly validated, establishing it as a key factor influencing slope stability (Ren et al. 2024). Slope is also a critical factor in landslide assessments. Studies have shown that the distribution of landslide points is highly consistent with slope, and the steeper the slope, the higher the probability of landslide occurrence (Tyagi et al. 2023). The impact of slope is widely recognized in landslide evaluation, and it is typically regarded as a core topographic factor determining slope stability (Hu and Yan 2024; Liu et al. 2024). NDVI, as an indicator of vegetation cover, is often strongly correlated with landslide stability. In areas with good vegetation cover, extreme rainfall or prolonged rainfall events may lead to deep soil saturation. When pore water pressure increases sharply, landslides may be triggered (Hsu 2023; Lizárraga and Buscarnera 2020; Stabile and Colângelo 2024). Different types of vegetation contribute differently to soil stability. Shallow-rooted vegetation primarily affects the stability of surface soils, while deep-rooted plants help consolidate deeper soils. If a landslide occurs in deeper soil layers that are beyond the reach of the root systems, landslides can still occur even in areas with abundant surface vegetation (Toprak et al. 2024; Zhao et al. 2021). Finally, aspect, as a topographic factor, also has an influence on landslide assessment (Dutta et al. 2024). Research has shown that slopes with different aspects respond differently to rainfall and other environmental conditions, which places aspect as a relevant

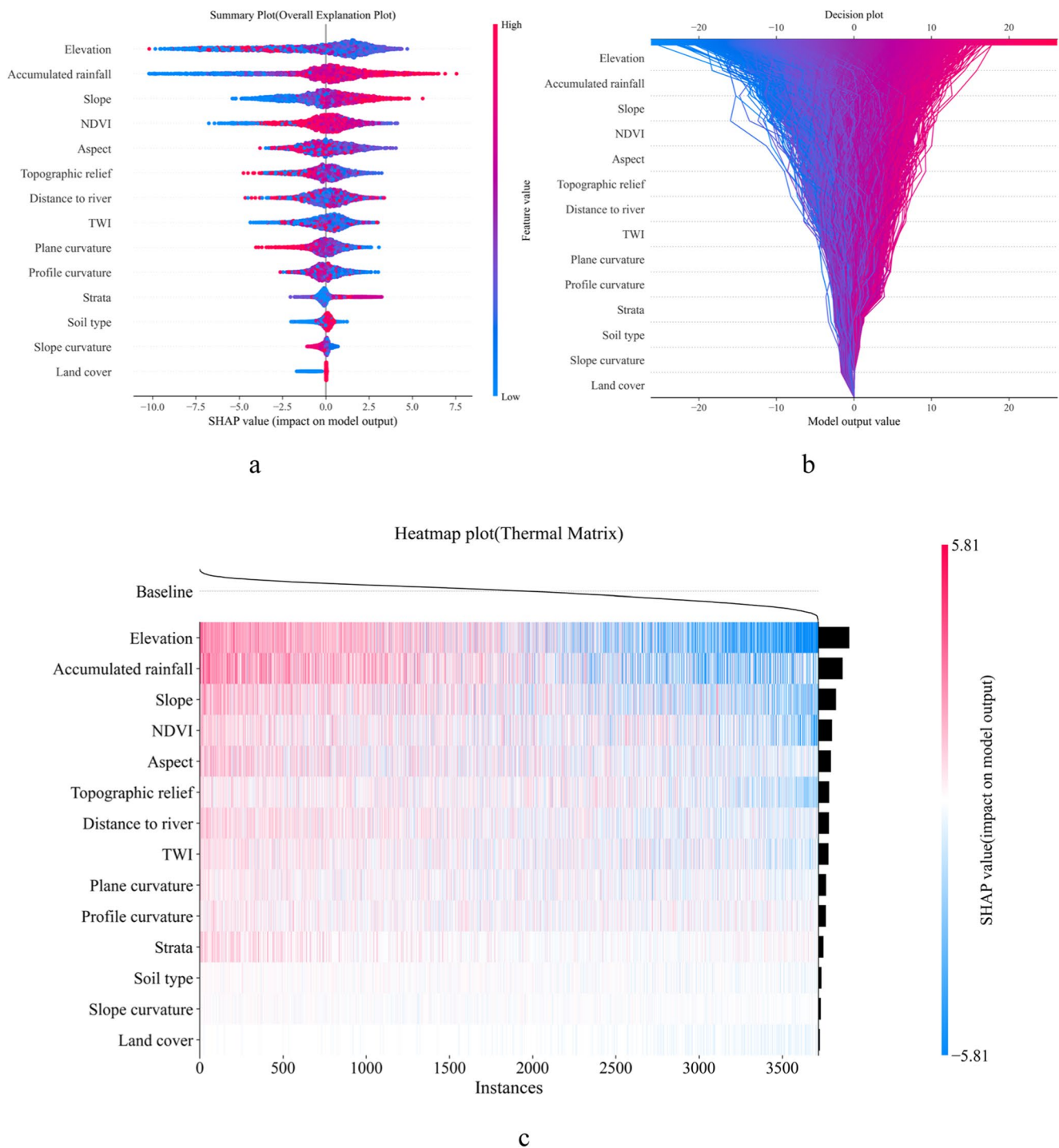


Fig. 17 SHAP value analysis based on the XGBoost model within the Autogluon framework. (a) Summary plot analysis; (b) decision plot analysis; (c) heatmap plot analysis

factor in landslide evaluation analysis (Kavzoglu et al. 2015). Landslide occurrence is the result of the combined effect of multiple factors, with elevation, accumulated rainfall, slope, NDVI, and aspect playing key roles in landslide assessments across different regions.

Conclusions

This paper, based on the extreme rainfall event in eastern Guangdong in August 2018, combined remote sensing imagery and 14 environmental factors to analyze

the spatial distribution of landslides and the multi-factor coupling mechanisms. Through remote sensing image interpretation, 6,194 landslide records were identified, with landslides primarily distributed in the southwestern, southeastern-central, and northeastern parts of the study area. The analysis revealed that the majority of landslides have an area between 1,000 and 2,000 m², which aligns with the clustering pattern of medium- to small-scale landslides. Landslides predominantly occurred at elevations between 74 and 374 m, with slopes ranging from 10° to 30°, facing southeast, south, and east, and TWI values between 4 and 6. The weathered granite layer was the primary triggering factor. High NDVI values (200–255), soil types (such as lateritic red soil), and terrain curvature were found to significantly influence landslide occurrence, with areas near rivers being more prone to triggering landslides. Through analysis using the optimal WeightedEnsemble model within the AutoGluon framework, it was confirmed that elevation, accumulated rainfall, slope, and NDVI are the most important triggering factors for landslides, with elevation having the greatest impact. High landslide occurrence areas were concentrated in regions with low to middle elevations, moderate slopes, dense vegetation, and high rainfall. The coupling of terrain and rainfall is identified as the key mechanism for landslide occurrence. Finally, based on the XGBoost model and SHAP analysis within the AutoGluon framework, it was confirmed that elevation, rainfall, and slope contribute the most to landslide occurrence, with their impact being decisive. The duration of rainfall may be the reason for the deviation of the landslide count and area from the conventional fitting curve in this event.

Acknowledgements All the authors who contributed to the study have approved the final version. We thank the anonymous reviewer and the editor, whose constructive and helpful comments improved this manuscript.

Author contributions Chenchen Xie: Data curation; Writing—original draft; Writing—review and editing; Validation. Chong Xu: Data curation; Methodology; Conceptualization; Writing—review and editing; Supervision; Formal analysis. Xiwei Xu: Conceptualization; Data curation; Supervision. Yuandong Huang: Data curation. Huiran Gao: Data curation.

Funding This research was funded by the State Grid Fundamental Research (No. 5700–202458226 A-1-1-ZN).

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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